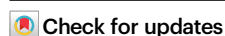


Enabling the synthesis of O3-type sodium anion-redox cathodes via atmosphere modulation

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Yixiao Qiu^{1,7}, Qinzhe Liu^{1,7}, Jiangwei Tao^{1,7}, Pu Yan^{1,2}, Guangsu Tan¹, Fanjun Liu³, Danyu Wang¹, Na Yu¹, Nian Zhang⁴, Yong Yang¹, Weiwei Wang⁴, Yuzhu Wang⁴, Kecheng Cao^{1,2}, Jingyang Wang⁵✉, Zhengyan Lun^{3,6}✉ & Chao Xu¹✉

The pursuit of advanced battery chemistries with enhanced energy density necessitates the exploration of new materials, a process intricately tied to synthesis science. Despite the promise of O3-type sodium oxygen anionic redox cathodes as high-capacity materials, their development has been severely hindered by a lack of understanding regarding synthetic mechanisms. Here, we elucidate the pivotal role of atmospheric conditions, particularly oxygen content, in the synthesis of such materials by synchronizing multiple *operando* characterization techniques to monitor changes in both solid and gaseous components. Utilizing the O3-Na[Li_{1/3}Mn_{2/3}]O₂ system as a model, we demonstrate that a low oxygen environment is essential and the reaction is highly complex as evidenced by multiple oxygen uptake and release processes, resulting in numerous intermediates. This behavior contrasts sharply with Na-Mn-O and Li-Mn-O ternary systems which show less significant oxygen dynamics, underscoring the unique reaction mechanism within the Na-Li-Mn-O system. We further adopt a dynamic controlled atmosphere approach to modulate oxygen concentration and demonstrate successful synthesis of Ti-substituted NaLi_{1/3}Mn_{2/3-x}Ti_xO₂ materials, all exhibiting capacities surpassing 190 mAh g⁻¹. Our findings highlight the importance of the atmospheric conditions for the synthesis of oxide cathode materials and these fundamental insights unlock avenues for developing novel high-energy-density sodium-ion battery chemistries.

Advancements in synthesis serves as a cornerstone for all facets of materials science and engineering. For instance, the controlled fabrication of nano-sized or even single-atom materials provides numerous opportunities for designing novel catalysis materials with improved

efficiency and selectivity^{1–3}. This progress is also evident in materials for electrochemical energy storage applications^{4–10}. The synthesis methods for layered cathode materials in lithium-ion batteries have continuously evolved as the nickel content increases to achieve higher

¹School of Physical Science and Technology, ShanghaiTech University, Shanghai, China. ²Shanghai Key Laboratory of High-resolution Electron Microscopy, ShanghaiTech University, Shanghai, China. ³School of Chemical Sciences, University of Chinese Academy of Sciences, Beijing, China. ⁴Shanghai Synchrotron Radiation Facility, Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai, China. ⁵School of Sustainable Energy and Resources, Nanjing University, Suzhou, China. ⁶Binzhou Institute of Technology, Weiqiao-UCAS Science and Technology Park, Binzhou, Shandong Province, China.

⁷These authors contributed equally: Yixiao Qiu, Qinzhe Liu, Jiangwei Tao. ✉e-mail: jy_wang@nju.edu.cn; zylun@ucas.ac.cn; xuchao1@shanghaitech.edu.cn

energy density^{11,12}. Specifically, the lithium precursor has changed from lithium carbonate to lithium hydroxide^{13,14}, and the calcination process has transitioned from single-step to multi-step^{9,15,16}, and importantly, the calcination atmosphere has adopted pure oxygen instead of air^{17,18}.

The evolutions of synthesis conditions are intricately guided by the thermodynamics and kinetics that govern the desired synthesis reactions, all aiming at achieving the production of a pure target product. An illustrative example involves the increase in the oxygen content in the calcination atmosphere as the nickel content increases in $\text{LiNi}_x\text{Mn}_y\text{Co}_{1-x-y}\text{O}_2$ (NMC)-type cathodes, leading to the oxidation state of nickel in the final lithiated product approaching trivalent¹⁷. To facilitate this reaction, a pure oxygen environment is imperative, ensuring a strong oxidizing environment that minimizes the presence of divalent nickel, which may lead to the formation of detrimental Li/Ni anti-site defects¹⁷. Additionally, the synergy between reaction thermodynamics and kinetics is evident in the synthesis of P2-type sodium transition metal oxides for application in sodium-ion batteries where the formation of various metastable species is observed, preceding the eventual attainment of thermodynamically stable compounds^{10,19}. Despite extensive efforts devoted to developing theoretical frameworks predicting emerging materials and their synthesis conditions, substantial challenges persist in synthesizing compounds predicted to be thermodynamically stable, whose formation pathways are intricately intertwined with diverse metastable compounds that are computationally difficult to be fully captured^{20–23}.

In this work, we use $\text{O3-Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$, a promising O3-type oxygen anionic redox (OAR) cathode material for sodium-ion batteries, to demonstrate the complex reaction mechanism in a ternary system featuring two alkaline metals, both of which can form various oxide species with manganese, and importantly, the pivotal role of the synthesis atmosphere in determining the reaction pathways during the synthesis. We highlight that, despite their attractive properties such as high capacity and good cycling stability demonstrated by $\text{Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ ²⁴, manganese-based O3-type OAR sodium cathodes are significantly scarcer^{21,24} as compared to the P2 polymorphs^{25–44}. The key issue lies in the lack of fundamental understanding into the synthesis reactions that leads to the inaccessibility of O3-type phase pure compounds. $\text{O3-NaLi}_{1/3}\text{Mn}_{2/3}\text{O}_2$ is predicted to be thermodynamically stable⁴⁵; however, Wang et al. demonstrated that using conventional atmosphere, namely argon, vacuum, air or oxygen, the product contains no or a limited amount of $\text{O3-Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ with various impurities, irrespective of the choice of lithium and sodium precursors and calcination temperature²⁴. The authors obtained high purity product only by purging the tube furnace with argon flow for 45 min, whereas a longer purging or continuous argon flow leads to impurities²⁴. These results clearly indicate the high complexity in the synthesis towards $\text{O3-Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ and the oxygen content in the atmosphere, essentially the oxygen chemical potential, is a critical factor that yet remains unexplored, resulting in limited progress in O3-type anionic redox sodium cathodes.

Here, we demonstrate that the high purity $\text{O3-Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ is accessible by fine-tuning the calcination atmosphere with 1–2% oxygen in argon (or nitrogen), and fully uncover how oxygen chemical potential influences the reaction mechanism through a series of *operando* characterization techniques. Our *operando* gas chromatography (GC) quantifies the dynamic oxygen release-uptake behavior during the synthesis, supported by intermediate compounds such as $\alpha\text{-NaMn(III)O}_2$ and $\text{Na}_3\text{Mn(V)O}_4$ captured by *operando* X-ray diffraction. The results demonstrate that the synthesis of $\text{O3-Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ requires a low oxygen content environment to oxidize initial low Mn-valence intermediates and subsequently avoid over-oxidizing to high Mn-valence impurities. This complex reaction scheme differs sharply from those of the two ternary systems, i.e., Na-Mn-O and Li-Mn-O, highlighting the unique

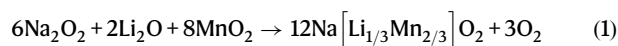
mechanism when both the two alkaline metal ions are present. Based on these fundamental insights, we propose a dynamic controlled atmosphere (DCA) approach starting with an oxidizing environment followed by an inert atmosphere. With this method, we successfully synthesize O3- $\text{Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ and O3-type Ti-substituted $\text{NaLi}_{1/3}\text{Mn}_{2/3-x}\text{Ti}_x\text{O}_2$ ($x = 1/18, 1/9$ and $1/6$) as high-capacity OAR cathode materials for sodium-ion batteries.

Results

Optimizing the calcination atmosphere

The honeycomb-ordered $\text{O3-Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ is isostructural to $\text{Li}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ (i.e., Li_2MnO_3), a classic oxygen-redox active cathode material, with the Li in the alkaline metal layer fully replaced by Na (Fig. 1a). First principles calculations (see methods for computational details) were first applied to identify proper composition for in-depth synthesis investigations by calculating the mixing enthalpy of structures with various degrees of Na-to-Li substitution. The results (Fig. 1b) show that the honeycomb-ordered $\text{Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ ($x = 0.75$, x in $\text{Na}_x\text{Li}_{1-x}\text{Mn}_{0.5}\text{O}_{1.5}$) has the lowest mixing enthalpy, indicating that it is thermodynamically favorable for O3-stacking structures, which is in good agreement with the previous report⁴⁵. We thereby utilized $\text{Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ for further synthesis investigations.

Initial attempts of the synthesis were conducted under the most conventional atmospheres used for the solid-state synthesis of layered oxide cathodes, namely pure oxygen, air and inert conditions such as argon. The choice of precursors, i.e., Na_2O_2 , Li_2O and MnO_2 , is adopted from the previous work by Wang et al., except for the use of Mn(IV)O_2 instead of Mn(III)O_2 due to the fact that the use of MnO_2 leads to purer products (Supplementary Fig. 1; Fig. 1c). The oxidation state of manganese, therefore, is +IV in both the precursor and final product. The proposed reaction (**reaction 1**) is oxygen-releasing, suggesting an inert atmosphere such as argon would be suitable to obtain the desired product. The corresponding XRD results (Fig. 1c), however, suggests otherwise.



The product obtained in Ar contains only a minor amount of a O3 phase, but instead is dominated by $\alpha\text{-Na}_{1-\delta}\text{MnO}_2$ and Li_2MnO_3 . The existence of $\alpha\text{-Na}_{1-\delta}\text{MnO}_2$ suggests that the atmosphere during the calcination is insufficient to sustain the Mn at +IV state. Unfortunately, simply increasing the oxygen content to 21% and 100% does not result in the formation of phase pure $\text{O3-Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$, instead, leads to various different impurities such as P3 structured phase ($\text{P3-Na}_x\text{Li}_y\text{Mn}_{1-y}\text{O}_2$ ($x < 1$), denoted as P3-NLMO; Fig. 1c). These results collectively show that the conventional gas environments are not able to directly produce the target $\text{O3-Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$. In specific, an inert atmosphere leads to impurities with lower-than-target Mn valence while highly oxidizing conditions favor the formation of P3-type phases. This also indicates that the synthesis of $\text{O3-Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ is sensitive to the calcination atmosphere, which requires dedicate tuning of the oxygen content.

We further selected three additional conditions, namely 1%, 2% and 5% oxygen in argon (or nitrogen), respectively, as the attempts above clearly indicate a moderate oxidizing environment is required. As shown in Fig. 1c, the three products are all dominated by the target $\text{O3-Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$. The diffraction signals arisen from the honeycomb superstructure is also clearly visible (in the 2θ range of 20° – 30°), in excellent agreement with the literature²⁴. The products synthesized with 1% O_2 and 2% O_2 only contain trace amounts of Li_2MnO_3 impurities, while 5% O_2 sample contains a small amount of a P3-NLMO phase. In-depth explorations of the reaction mechanisms are required to understand the critical role of oxygen content during the synthesis.

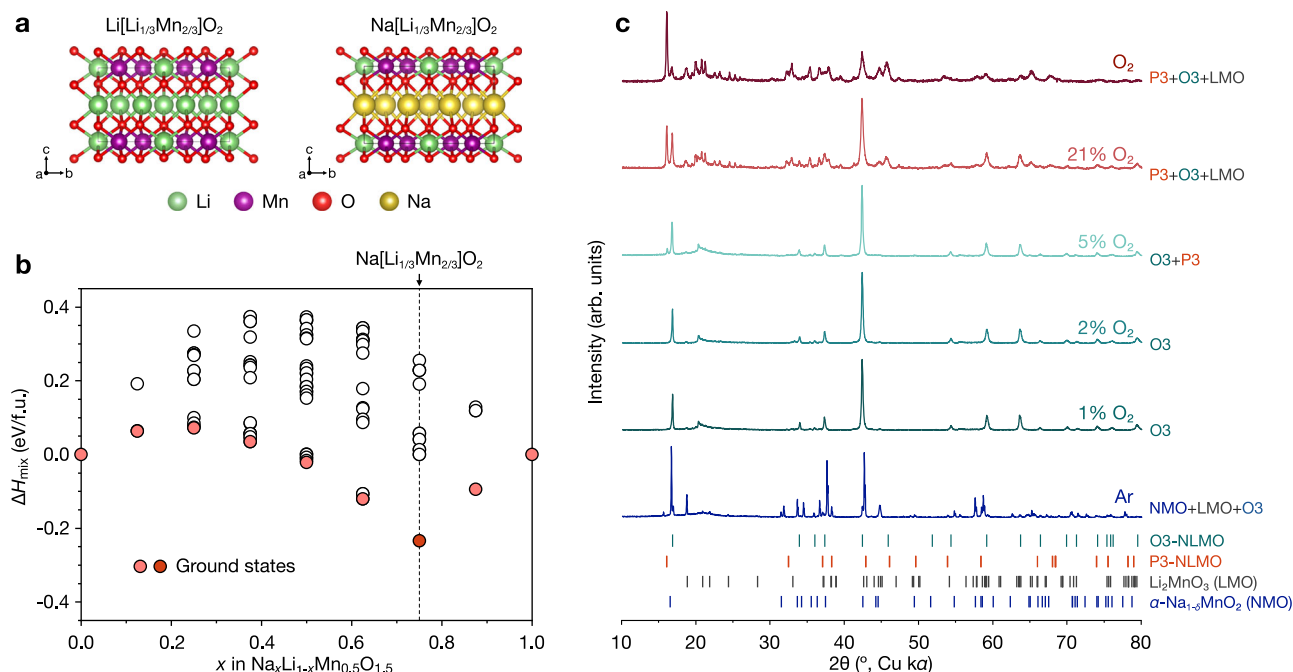


Fig. 1 | Optimizing the calcination atmosphere. **a** Illustration of the honeycomb-ordered $\text{Li}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ and $\text{Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ structure. Structural model, $\text{C2}/m$. **b** Formation energies of mixing enthalpy for $\text{Na}_x\text{Li}_{1-x}\text{Mn}_{0.5}\text{O}_{1.5}$. The vertical dashed line indicates the $\text{NaLi}_{1/3}\text{Mn}_{2/3}\text{O}_2$ stoichiometry where x is 0.75. The ground states at each Na content are highlighted as red filled circles. **c** XRD patterns of the synthesis

products under atmospheres containing various oxygen contents, namely 0% (pure Ar), 1%, 2%, 5%, 21%, and 100% (pure O_2), respectively. Note the 0%, 21% and 100% O_2 samples contain additional impurities which cannot be indexed by major structural databases. Source data are provided as a Source Data file.

Operando studies of $\text{O3-Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ synthesis during temperature ramping

We combine three *operando* techniques, including XRD, thermal gravimetric analysis (TGA) and gas chromatography (GC) to track the solid and the gaseous components evolution during the synthesis process. The *operando* GC is developed in-house, and is able to accurately quantify the oxygen content in the gas flow with a temporal resolution of ~2 min (Supplementary Fig. 2; Methods). Overall, the combined *operando* results (Fig. 2) of the synthesis conducted under 2% O_2 (in argon or nitrogen) show that the reaction is highly complex featured by a series of structural changes, companied by distinct oxygen release and uptake. For clarity, we divide the temperature ramping section into three reaction phases (Fig. 2a), primarily based on whether it is oxygen releasing or uptake, and discuss in details below.

Phase I: room temperature (RT) to -290°C ; oxygen release. The diffraction signals from the precursors, i.e., Na_2O_2 , Li_2O and MnO_2 are clearly observed at the beginning of the ramping process (Fig. 2, Supplementary Fig. 3). When the temperature reaches -160°C , a $\text{P3-Na}_x\text{Li}_y\text{Mn}_{1-y}\text{O}_2$ phase is formed but quickly disappears as the temperature continues to increase, forming $\alpha\text{-Na}_{1-\delta}\text{MnO}_2$ (O'3-NMO, Supplementary Fig. 3). Simultaneously, a large amount of oxygen is released during this phase, as evidenced by *operando* GC results (Fig. 2c) and the drastic weight loss in the TGA (Fig. 2d). Note that the amount of released oxygen and the weight loss is significantly higher than that of the Na_2O_2 disproportionation reaction alone, as supported by both the GC (theoretical: $2500\ \mu\text{mol}$; experimental: $3341\ \mu\text{mol}$) and TGA (theoretical weight loss: 7.84%; experimental: 10.03%) results (Supplementary Fig. 4). The extra gas release and weight loss can be attributed to the oxygen loss from the reaction between $\text{NaO}_x/\text{LiO}_x$ and MnO_2 because lower Mn valence compounds such as $\alpha\text{-Na}_{1-\delta}\text{MnO}_2$ is formed. This is particularly important as it indicates the nominal oxidation state of manganese is now lower than +IV by the end of phase I, and in order to return back +IV as in

the $\text{O3-Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$, an oxidative environment is required for the rest of the calcination process.

Phase II: -290°C to -550°C ; oxygen uptake. As the temperature increases, the amount of $\alpha\text{-Na}_{1-\delta}\text{MnO}_2$ decreases as evidenced by the reduced intensities of the diffraction signals. Note that $\alpha\text{-Na}_{1-\delta}\text{MnO}_2$ is still present at these temperatures as confirmed by ex situ high-resolution synchrotron X-ray diffraction results (Supplementary Fig. 5). Concurrently, $x\text{-Na}_3\text{MnO}_4$ forms, which denotes one or multiple polymorphs of Na_3MnO_4 that are structurally unknown as it cannot be indexed by major structural databases (see Supplementary Note 1 and Supplementary Figs. 6–8 for a detailed discussion of this assignment). Upon further increasing the temperature beyond 450°C , $x\text{-Na}_3\text{MnO}_4$ rapidly transforms to $\gamma\text{-Na}_3\text{MnO}_4$ (Fig. 2b, Supplementary Fig. 5). Lithium manganese compounds (Spinel-like) are observed at the early stage of II (Supplementary Fig. 5). During phase II, obvious oxygen uptake is observed (Fig. 2c) and the weight of the solid gradually increases (Fig. 2d), in agreement with the presence of more oxidized Mn compounds as compared to those at the end of phase I. Note that one discrepancy occurs at -390°C , i.e., an oxygen uptake signal is detected while no obvious mass increase is observed, which we attribute to a H_2O -release reaction (Supplementary Fig. 4). This issue does not affect the qualitative trend obtained by the *operando* GC and we performed additional experiments using anhydrous sodium hydroxide instead of highly hygroscopic Na_2O_2 and the results confirm that the nominal manganese valence initially decreases (phase I) but increases to above +IV after the oxygen-consuming phase II (Supplementary Fig. 9).

Phase III: -550°C to 700°C ; oxygen release. During this phase, $\gamma\text{-Na}_3\text{MnO}_4$, $\alpha\text{-Na}_{1-\delta}\text{MnO}_2$ and Li_2MnO_3 are identified as the key intermediates that further react to form the target product $\text{O3-Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ (Fig. 2b, Supplementary Note 2 and Supplementary Figs. 10–11), based on the *operando* PXRD and ex situ synchrotron PXRD. Therefore, assuming $\alpha\text{-Na}_{1-\delta}\text{MnO}_2$ is stoichiometric (i.e., $\delta = 0$), the reaction (reaction 2) toward $\text{O3-Na}[\text{Li}_{1/3}\text{Mn}_{2/3}]\text{O}_2$ can be

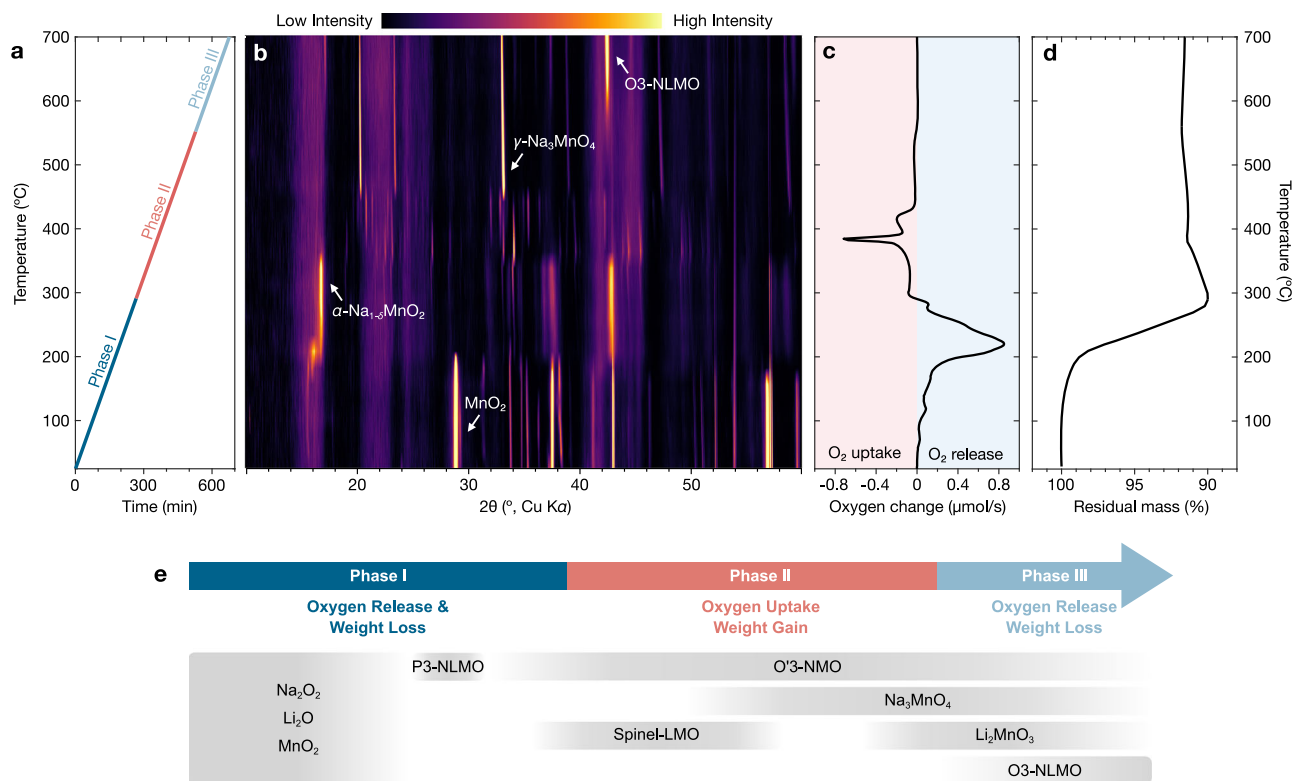
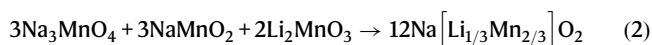


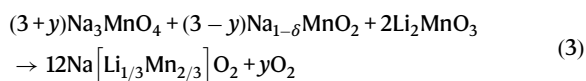
Fig. 2 | Operando tracking of the O3-Na[Li_{1/3}Mn_{2/3}]O₂ synthesis. **a** The temperature profile, **b** operando XRD, **c** operando GC and **d** TGA results during the temperature ramping of the O3-Na[Li_{1/3}Mn_{2/3}]O₂ synthesis under 2% O₂. **e** A

summary of the reaction processes and intermediate compounds during the temperature ramping. Source data are provided as a Source Data file.

described as follows:



Based on this mechanism, no change in the solid mass nor oxygen release or uptake is expected during this phase. However, TGA and operando GC results reveal that this reaction is oxygen releasing as evidenced by a slight decrease in the sample mass (91.77% to 91.56%; Fig. 2d) and an increase in the cumulative amount of oxygen released (1946.8 μmol to 2151.5 μmol; Supplementary Fig. 4a). This discrepancy can be explained by the excessive formation of Na₃MnO₄, which is balanced by presence of Na-deficient α-Na_{1-δ}MnO₂ to maintain the overall Na:Mn ratio, thereby raising the nominal oxidation state of manganese to higher than that +IV (Supplementary Fig. 9). Based on this understanding, we propose the following generalized reaction (reaction 3) which accounts for this variation:



Note that the values of y and δ are interrelated by the total amount of sodium, and they further determine the nominal Mn valence of the mixture of α-Na_{1-δ}MnO₂, Li₂MnO₃ and Na₃MnO₄.

Collectively, the results from operando XRD, GC and TGA clearly show that the reactions occur during the synthesis of O3-Na[Li_{1/3}Mn_{2/3}]O₂ are highly complex, featured by the formation of multiple Na-Mn-O and Li-Mn-O intermediate phases as well as distinct oxygen uptake and release, demonstrating that this process is sensitive to the atmospheric condition, i.e., the oxygen chemical potential (Fig. 2e). Notably, our results unambiguously explain why calcinations conducted using conventional conditions, no matter inert

atmosphere such as argon or nitrogen, air, or pure oxygen, lead to various impurities in the final product: (1) for an inert environment, the manganese is reduced at low temperature (phase I) and cannot be oxidized back to +IV with low oxygen chemical potential at higher temperature, resulting in low-valence impurities such as α-Na_{1-δ}MnO₂; (2) for strong oxidizing environments such as air and pure oxygen, high valence impurities such as Na₃MnO₄ exhibit preferential formation thermodynamically.

Operando studies of Na-Mn-O and Li-Mn-O ternary systems during ramping

To further understand the independent phase evolution upon heating in Na-Mn-O and Li-Mn-O ternary systems and the interplay between them, operando experiments were also conducted using the same precursor sets: Na₂O₂ + MnO₂, or Li₂O + MnO₂, respectively, under the same atmosphere of 2% O₂ in argon. The molar ratio of the reactants is maintained the same as that in the NLMO synthesis, i.e., Na₂O₂:MnO₂ = 3:4 and Li₂O:MnO₂ = 1:4, respectively. For the Na-Mn-O system, the operando XRD results (Fig. 3a, b) show that multiple phase changes occur, leading to the formation of α-Na_{1-δ}MnO₂ and then Na₃MnO₄ as the temperature increases to 700 °C, which is in close similarities to the observation in the NLMO synthesis (Supplementary Fig. 12), suggesting that the reaction occur during the NLMO synthesis is predominantly governed by Na-Mn-O. The unidentified species visible in phase II is also presented upon heating the ternary Na-Mn-O system, indicating that these are some unknown Na-Mn-O materials. For the Li-Mn-O system, spinel Li_xMnO_y also starts to form above ~300 °C and persists till 700 °C (a minor amount of Mn₂O₃ is produced between ~500 °C and 650 °C; Fig. 3d, e), which is different from the Na-Li-Mn-O quaternary system where Li₂MnO₃ is formed at high temperatures. We propose that the formation of various Na-Mn-O species with different Na:Mn stoichiometry results in a dynamic Li:Mn ratio,

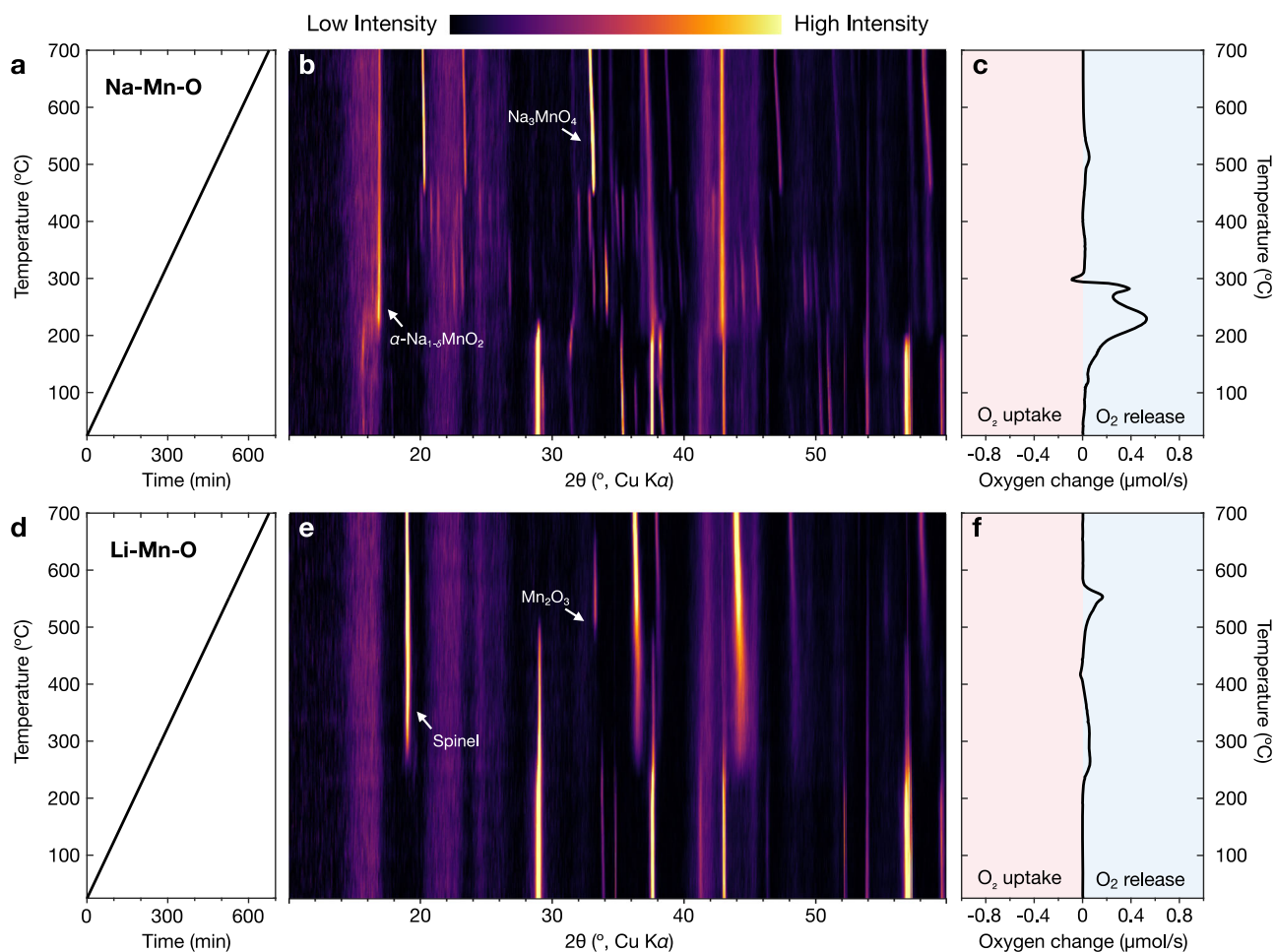


Fig. 3 | Operando characterizations of Na-Mn-O and Li-Mn-O ternary systems. **a** The temperature profile, **b** operando XRD and **c** operando GC results of the Na-Mn-O system during the temperature ramping process. **d** The temperature profile, **e** operando XRD and **f** operando GC results of the Li-Mn-O system during the

temperature ramping process. The calcination for both systems were carried out under a composition fixed atmosphere containing 2% O₂. Source data are provided as a Source Data file.

thus continuously alters the reaction thermodynamics for the Li-Mn-O ternary system along the calcination process, leading to the formation of different Li-Mn-O compounds.

Such intricate interplays also strongly impact the oxygen uptake/release behavior, which exhibits a stark contrast between the Na-Li-Mn-O quaternary system and the two ternary systems (Fig. 3c, f). For the Na-Mn-O system, similar oxygen releasing processes associated with peroxide disproportionation and formation of α -Na_{1-x}MnO₂ are observed at the beginning of the calcination. However, unlike the presence of phase II (~280 °C to ~550 °C) for the NLMO during which a significant amount of oxygen is consumed, minimum oxygen uptake is observed during the ramping processes for both of the two ternary systems. Based on the fact that strong α -Na_{1-x}MnO₂ diffraction signals persist in the Na-Mn-O system (Fig. 3b) while their intensity drops significantly in the Na-Li-Mn-O system (Fig. 2b) after the first oxygen-uptake spike at 390 °C in NLMO (Fig. 2c), we argue that this oxygen-uptake process is closely related to the oxidation of α -Na_{1-x}MnO₂ to a new phase with a higher Mn valence, which is a crucial step to form O3-Na[Li_{1/3}Mn_{2/3}]O₂ in the Na-Li-Mn-O system. These results show that the reaction thermodynamics of the two ternary systems, Na-Mn-O and Li-Mn-O, strongly affect each other, leading to the dynamic change in the oxygen release and uptake behavior, which plays an essential role in the phase evolution to form O3-Na[Li_{1/3}Mn_{2/3}]O₂.

Impact of oxygen content and dynamic controlled atmosphere

To unravel the impact of oxygen chemical potential on the reaction thermodynamics and kinetics, we track the environmental oxygen content evolution (using operando GC) upon calcination under three different atmosphere conditions, i.e., 1%, 2% and 5% oxygen in Ar/N₂, respectively. All the three reactions exhibit a similar qualitative and quantitative evolution till ~400 °C, by which oxygen release associated with the disproportionation of peroxide and the formation of reduced manganese compounds (phase I) occur followed by a subsequent oxygen uptake (Fig. 4a, Supplementary Fig. 13). The variation in the synthesis atmosphere primarily leads to different oxygen uptake/release behavior from ~400 °C and onwards where a higher O₂ environment exhibits more oxygen uptake during the remaining phase II and less oxygen release during phase III. Thus, a constant oxidizing environment throughout the calcination process can result in over-oxidation of Mn-containing species, preventing the formation of a phase-pure O3-Na[Li_{1/3}Mn_{2/3}]O₂.

Based on the generalized reaction (3), we further investigate the thermodynamic influence of the oxygen partial pressure through theoretical calculations of the reaction energy (Supplementary Table 1). The results (Supplementary Note 3 and Supplementary Fig. 14) clearly indicate that decreasing the oxygen partial pressure lowers the reaction energy, thereby favoring the formation of

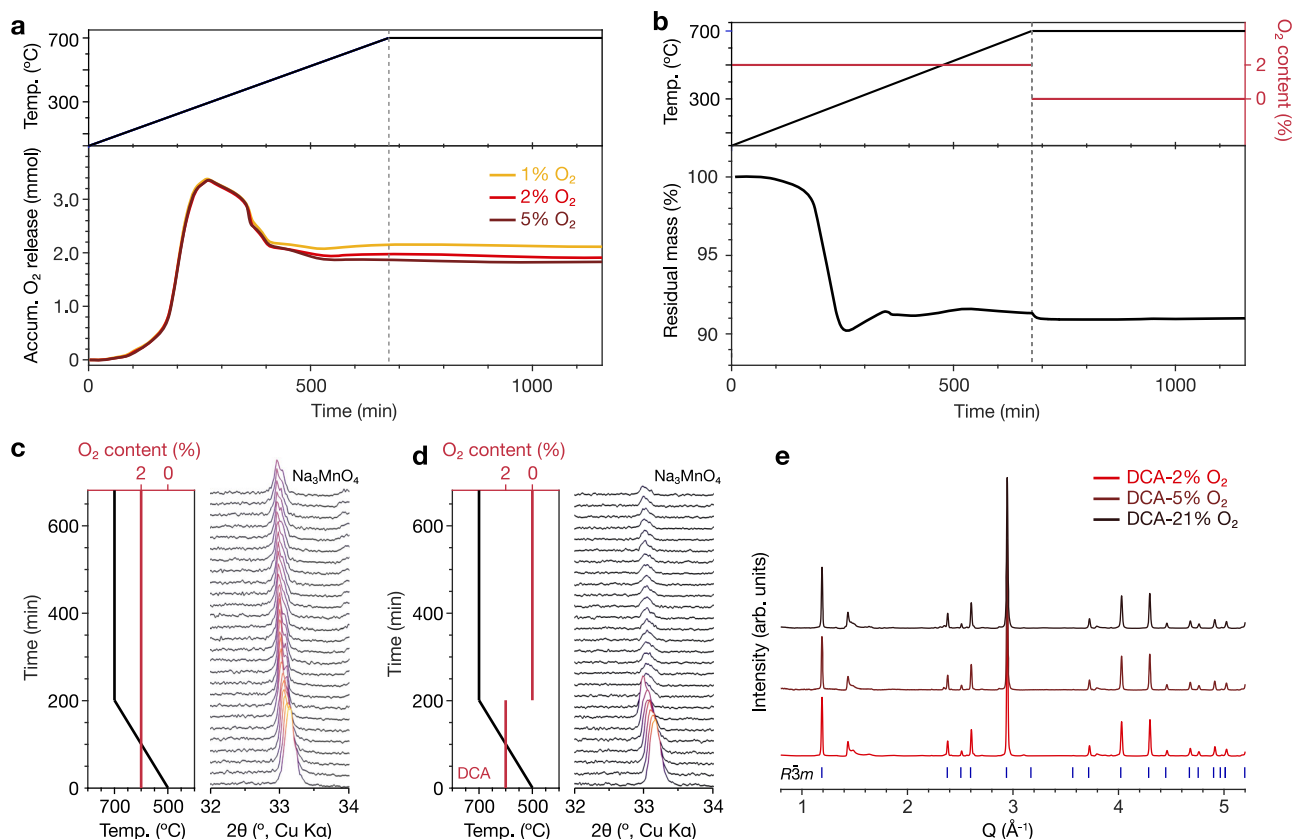


Fig. 4 | Effect of oxygen content. **a** Cumulative oxygen release using composition fixed atmosphere containing 1%, 2% and 5% O₂, respectively. **b** TGA result of the synthesis using dynamic controlled atmosphere approach, which includes a 2% O₂ flow during the ramping process and a pure nitrogen flow during the temperature hold. Bragg peak (Na₃MnO₄) evolution during the calcination using composition

fixed atmosphere (2% O₂) (**c**) and DCA approach (**d**). **e** XRD results of O3-NLMO synthesized via dynamically controlled atmosphere containing 2%, 5% and 25% O₂ for the ramping process, respectively. Source data are provided as a Source Data file.

O3-NLMO. In addition, a less Na-deficient α -Na_{1.6}MnO₂ requires a significantly lower oxygen partial pressure to achieve the same reaction energy when comparing Na_{0.83}MnO₂ to Na_{0.75}MnO₂. Guided by these theoretical insights, a less oxidizing or inert environment is recommended for phase III during the ramping and the subsequent high temperature holding at 700 °C in order to alter the reaction thermodynamics, favoring the formation of phase-pure O3-NLMO. We therefore adopt a DCA approach which contains oxygen for the ramping process but switches to inert atmosphere such as argon or nitrogen at high temperatures. Indeed, the TGA result clearly shows a sudden mass drop (-0.4%; Fig. 4b) when the atmosphere is switched to nitrogen (controlled by built-in mass flow controllers), which is not observed when a constant 2% O₂ flow is supplied (Fig. 2a). This mass loss is attributed to oxygen release as a result of the reduction of high valence Mn impurities such as Na₃MnO₄ (Fig. 4c, d). For reaction with fixed environmental oxygen content, only slight decrease in intensity (Fig. 4c), suggesting that majority of Na₃MnO₄ impurity remains unreacted. In contrast, a rapid conversion of Na₃MnO₄ impurity at high temperature can be identified after switching the atmosphere from 2% O₂ to pure nitrogen in the DCA synthesis as suggested by the attenuation of corresponding Bragg peaks (Fig. 4d). Note that distinct peak shifts to lower angles were observed in both Fig. 4c, d during the temperature ramping (Supplementary Fig. 15), which are attributed to the thermal expansion of Na₃MnO₄. Moreover, we also applied other oxygen content during the ramping, 5% and 21% to test the generality of our method. The 21% is selected to simulate the air condition. Note that using fixed (in composition) gas flow (Fig. 1c), the product contains impurities such as P3-NLMO. By dynamically switching to N₂ at high temperature, the formation of O3-Na[Li_{1/3}

₃Mn_{2/3}]O₂ can be clearly promoted as a result of Mn reduction in an inert atmosphere as shown in Fig. 4e. Only a minimum amount of Li₂MnO₃ and Na₃MnO₄ remains in the DCA-5% and DCA-21% O₂ samples, which may be eliminated by extending the duration of temperature holding.

O3-type titanium-substituted NaLi_{1/3}Mn_{2/3-x}Ti_xO₂ cathode materials

The DCA approach we proposed by systematic tracking of the phase, weight, and gas environment evolution upon solid-state synthesis can be highly expandable. As a demonstration, three O3-type titanium-substituted NaLi_{1/3}Mn_{2/3-x}Ti_xO₂ ($x = 1/18, 1/9$ and $1/6$) oxygen redox cathode materials were synthesized using the same method. The ramping process during the calcination was carried out under 2% O₂ environment whereas the constant-temperature and cooling steps are under pure argon, i.e., DCA-2% O₂. Similar to the synthesis of NLMO, the DCA approach suppresses the formation of Li₂MnO₃ and Na₃MnO₄ impurities (Supplementary Fig. 16), as confirmed by the XRD results (Fig. 5a for NLMT0-1/6; NLMT0-1/18 and NLMT0-1/9 are shown in Supplementary Fig. 11). The lattice parameters for the NLMTOs (based on the C2/m structural model) are summarized in Fig. 5b (more details in Supplementary Fig. 17 and Supplementary Table 2; Supplementary Fig. 18 and Supplementary Table 3 for refinement results based on the R3m structure). Notably, the *a*, *b* and *c* lattice parameters gradually increase as the Ti⁴⁺ content increases, which is ascribed to its larger ionic radius (0.605 Å) compared to that of the Mn⁴⁺ (0.53 Å), indicating successfully incorporation of Ti into the structure (stoichiometries confirmed by inductively coupled plasma-optical emission spectroscopy; Supplementary Table 4).

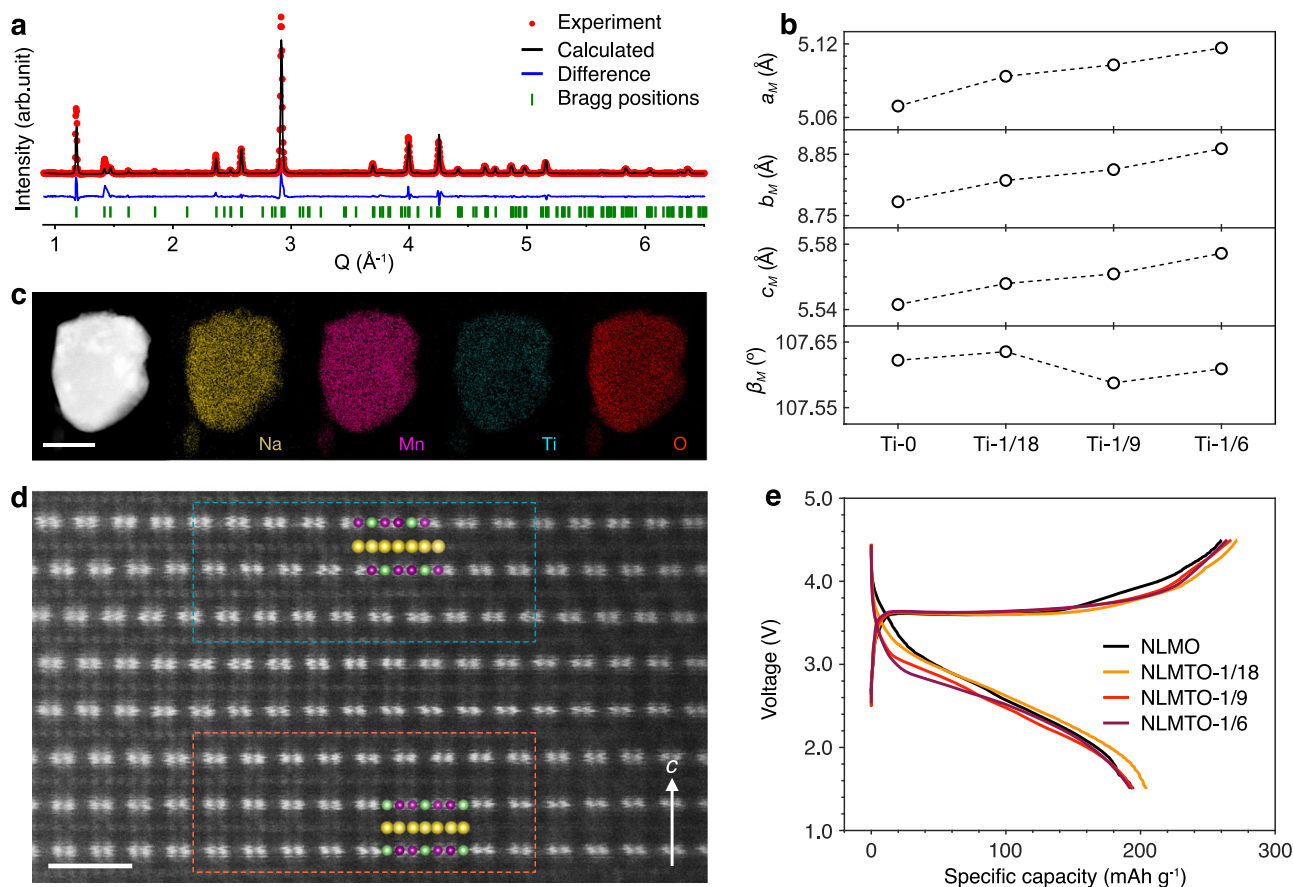


Fig. 5 | Structural and electrochemical properties of $\text{NaLi}_{1/3}\text{Mn}_{2/3-x}\text{Ti}_x\text{O}_2$ ($x = 1/18, 1/9$ and $1/6$). **a** XRD pattern and Rietveld refinement of the pristine NLMT0-1/6. **b** Lattice parameters of $\text{NaLi}_{1/3}\text{Mn}_{2/3}\text{O}_2$ and $\text{NaLi}_{1/3}\text{Mn}_{2/3-x}\text{Ti}_x\text{O}_2$ ($x = 1/18, 1/9$ and $1/6$) from Rietveld refinements. Structural model, $C2/m$. **c** SEM-EDX elemental mapping of the pristine NLMT0-1/6. Scale bar, $1\ \mu\text{m}$. **d** HAADF-STEM image of the pristine NLMT0-1/6. Scale bar, $1\ \text{nm}$. Red and blue rectangles illustrate monoclinic

unit cells viewed from the $[100]$ and $[110]$ zone axes, respectively. Yellow, green, purple circles represent sodium, lithium, and transition metals (manganese and titanium), respectively. **e** Voltage profiles of NLMO and NLMT0s during the first cycle at a rate of $C/20$ between 1.5 and $4.5\ \text{V}$ (i.e., $9.5\ \text{mA/g}$; $1\ \text{C} = 190\ \text{mA/g}$). Source data are provided as a Source Data file.

The as-synthesized NLMT0-1/6 has a plate-like morphology with a particle size of $\sim 2\ \mu\text{m}$ as shown in the scanning electron microscope (SEM) images (Supplementary Fig. 19), and the energy-dispersive spectroscopy (EDS) mappings confirm that Na, Mn, Ti, and O are uniformly distributed in the particle (Fig. 5c). The aberration-corrected high-angle annular dark field scanning transmission electron microscopy (HAADF-STEM) image (Fig. 5d) reveals the typical layered structure with dumbbell-like features arisen from the honeycomb ordering⁴⁶. The compound also presents various stacking faults, which can be attributed to rotational⁴⁷ and/or transitional stacking faults (Supplementary Note 4, Supplementary Figs. 20, 21). The degree of such stacking faults is estimated to be approximately 18% using FAULTS (Supplementary Table 5, Supplementary Fig. 22), which propagate along the c axis (Fig. 5d, Supplementary Fig. 22). The distances between adjacent TM layers are measured to range from $5.39\ \text{\AA}$ to $5.56\ \text{\AA}$ (Supplementary Fig. 23), which is close to value determined from lattice parameters from the refinement ($c_M \times \sin(\beta_M)$, $5.31\ \text{\AA}$; Fig. 5b). The discrepancy is attributed to sample preparation, including possible lattice strain from grinding and de-sodiation from sonication in ethanol, as well as potential measurement errors and lattice distortion due to electron beam irradiation. The line-scan electron energy loss spectroscopy (EELS) measurement along the transition metal layer confirms the presence of Ti and Mn in the TM layer and the Mn L-edge X-ray absorption spectra confirm the Mn in NLMT0-1/6 is close to Mn(IV) state (Supplementary Fig. 24).

The electrochemical performance of NLMO and $\text{NaLi}_{1/3}\text{Mn}_{2/3-x}\text{Ti}_x\text{O}_2$ ($x = 1/18, 1/9$ and $1/6$) synthesized using DCA-2% O_2 approach was investigated in sodium half cells and the corresponding first-cycle voltage profiles of the four materials are shown in Fig. 5e. All four charging curves display a prolonged plateau at approximately $3.6\ \text{V}$, succeeded by a sloping region, while the discharging curves exhibit a slope shape. Interestingly, the titanium substitution demonstrates a noticeable impact on the charge and discharge capacities: a minor substitution of $1/18$ initially enhances the capacity, whereas further increases to $1/9$ and $1/6$ result in lower capacities. The NLMT0-1/18 sample exhibits the highest charge and discharge capacities of $271.8\ \text{mAh/g}$ and $204.3\ \text{mAh/g}$, respectively. Upon increasing the rate to $0.2\ \text{C}$ ($38\ \text{mA g}^{-1}$), the discharge capacities for NLMO, NLMT0-1/18, NLMT0-1/9, and NLMT0-1/6 are $160.1\ \text{mAh g}^{-1}$, $159.9\ \text{mAh g}^{-1}$, $138.0\ \text{mAh g}^{-1}$, and $102.4\ \text{mAh g}^{-1}$, respectively, during the first $0.2\ \text{C}$ ($38\ \text{mA g}^{-1}$) cycles (Supplementary Fig. 25), further demonstrating that excessive cooperation of Ti leads to lower reversible capacities. These results indicate that higher levels of Ti substitution lead to lower reversible capacities, which can be attributed to the electrochemically inert nature of Ti^{4+} . However, the Ti-substituted samples exhibit a significantly improved capacity retention over long-term aging compared to the NLMO cathode. Among the Ti-substituted samples, NLMT0-1/18 demonstrated the best overall performance, delivering a similar capacity at $0.2\ \text{C}$ ($38\ \text{mA g}^{-1}$) compared to the unsubstituted NLMO while significantly improving capacity retention. This

improvement is tentatively attributed to an enhanced reversibility of sodiation due to suppressed parasitic reactions related to oxygen redox as well as Jahn-Teller distortion of manganese, as seen in Ti-substituted P2-type anionic redox cathode materials⁴⁸. Nonetheless, further comprehensive investigation into the electrochemical performance and structural evolution during sodiation and de-sodiation is required, although it lies beyond the scope of this study.

Discussion

In this work, we have decisively unraveled the intricate reaction mechanism underlying O3-NLMO synthesis and extended this understanding to the successful creation of a series of novel Ti-substituted O3-NLMOs. Through fine adjustment of oxygen content in the calcination atmosphere, we have demonstrated the feasibility of synthesizing O3-NLMO under low oxygen environments. Employing a combination of *operando* characterization techniques, including *operando* XRD, GC and TGA, we have shown a remarkably complex reaction scheme characterized by multiple intermediates and distinct oxygen dynamics of release and uptake. These findings, distinct from those of Na-Mn-O and Li-Mn-O ternary systems, underscore the pivotal role of oxygen in governing the thermodynamics and kinetics of quaternary compound synthesis. Importantly, we have pioneered a DCA approach to manipulate reaction dynamics, favoring the formation of O3-type oxide cathode materials, and have successfully synthesized three NLMOs which all have high reversible capacities exceeding 190 mAh g⁻¹.

Our work creates a versatile synthesis approach for high-capacity oxide cathode materials by advancing conventional synthesis conditions, thereby propelling advancements in sodium-ion battery research and development. Furthermore, the methodology we have developed for characterizing solid-state synthesis, which tracks the evolution of both solid and gaseous components concurrently, is indispensable for comprehensively unraveling reaction mechanisms.

Methods

Theoretical calculations

First-principles calculations were conducted on various Li-substituted Na₂MnO₃ structures using the PWmat code^{49,50} to ascertain the mixing enthalpy in each instance. The generalized gradient approximation (GGA) with Hubbard-type U correction and NCPP-SG15-PBE Pseudopotential were employed, significantly mitigating calculation errors. The Monkhorst-Pack method with 6 × 3 × 6 meshes was utilized for sampling the k-point in the reciprocal space. The conjugate gradient optimization method was employed with convergence criteria of 0.01 eV Å⁻¹ for atomic forces and 0.01 eV N⁻¹ per atom for lattice stresses. The plane wave cutoff energy for wavefunction (ECUT) and the cutoff energies for the soft charge density and the potential (ECUT2) were established at 70 Ryd (1 Ryd = 13.6057 eV) and 280 Ryd. This configuration ensures optimal lattice refinement and adequate relaxation of the calculated structures, given that relaxation calculations necessitate the inclusion of atomic forces. For the optimization of phases with different Na:Li ratios, Na₂MnO₃ (C2/m space group) was used as the initial base structure and all the Na were replaced by randomly distributed Li and Na while the Mn sites were fixed. The various ordered phases were obtained by excluding duplicates based on their symmetry. The enthalpies of the mixtures, characterized by various substitution structures, cannot be directly derived from simulation calculations. Instead, they must be obtained through the following equation:

$$\Delta H_{\text{mix}} = E_{\text{Li}_{1-x}\text{Na}_x\text{MnO}_{3.5}} - xE_{\text{NaMnO}_{3.5}} - (1-x)E_{\text{LiMnO}_{3.5}}$$

where E denotes the total energy of the system. By comparing the magnitudes of this thermodynamic quantity, it is possible to gauge the thermodynamic stability of the structure and find out the most stable

phase in different substitution cases through the comparison of their magnitudes.

Reaction energies of NLMO (Na₃LiMn₂O₆) formation at different oxygen partial pressure were calculated based on the formation energies of each reactant and product. The method for calculating the formation energies is as follows: The system energies of Li₂MnO₃, Na₃MnO₄, Na_xMnO₂ ($x = 5/6, 3/4$) and Na₃LiMn₂O₆ are calculated by using the RELAX mode in PWmat code^{49,50}, with the same parameters used for calculating the mixture enthalpy. The GGA + U function and the NCPP-SG15-PBE pseudopotential were used. The conjugate gradient optimization method was employed with convergence criteria of 0.01 eV Å⁻¹ for atomic forces and 0.01 eV N⁻¹ per atom for lattice stresses, and the plane wave cutoff energies and soft charge density and potential cutoff energies are established at 70 Ryd and 280 Ryd. The system energy of Na_xMnO₂ is calculated by calculating all possible Na/vacancy orderings with the GGA + U function and selecting the lowest energy structure for consideration. Then, the formation energy of each structure can be obtained by subtracting the total energy of each atom from the calculated system energy. The contribution from the oxygen partial pressure is accounted for by computing the oxygen chemical potential, μ_{O_2} , based on the equation below^{10,51}:

$$\mu_{\text{O}_2}(T, p_{\text{O}_2}) = h_{\text{O}_2}(T, p_0) - T \left(s_{\text{O}_2}(T, p_0) - k_B \times \ln \left(\frac{p_{\text{O}_2}}{p_0} \right) \right)$$

where p_0 is the reference pressure, p_{O_2} is the oxygen partial pressure and k_B is the Boltzmann constant. h_{O_2} and s_{O_2} represent the enthalpy and entropy oxygen gas per molecule, respectively.

Material synthesis

The materials were synthesized through conventional solid-state methods. For O3-NaLi_{1/3}Mn_{2/3}O₃ (denoted as O3-NLMO), stoichiometric amounts of Na₂O₂ (Sigma-Aldrich, ≥95%), Li₂O (Sinopharm Chemical Reagent Co., Ltd. ≥99.99%), MnO₂ (Aladdin, ≥99%) precursors were first mixed in an agate mortar by hand thoroughly in an argon glovebox. The Na₂O₂ was received in the form of granular beads (~2 mm in diameter), which were crushed, ground and sieved (300 mesh) to reduce the particle size, facilitating better contact with other precursors. Note these processes involving handling Na₂O₂ often led to a decrease in the moisture level in the argon glovebox (<0.1 ppm H₂O), indicating its high hygroscopicity. The MnO₂ was dried in a vacuum oven at 120 °C for 12 h. The mixture was calcined at 700 °C for 8 h with a temperature ramping and cooling rate of 1 °C/min using a variety of atmosphere conditions (including pure argon/nitrogen and oxygen; 1%, 2%, 5% and 21% oxygen in argon/nitrogen). For NaLi_{1/3}Mn_{2/3-x}Ti_xO₂ (denoted as O3-NLMO), stoichiometric precursors of Na₂O₂ (Sigma-Aldrich, ≥95%), Li₂O (Sinopharm Chemical Reagent Co., Ltd. ≥99%), MnO₂ (Aladdin, ≥99%), TiO₂ (Aladdin, ≥99.5%, 60 nanometer, rutile) were thoroughly mixed in an agate mortar by hand in an argon glovebox, followed by calcination at 800 °C for 8 h and cooled to room temperature (1 °C/min for heating and cooling) in a 2% oxygen atmosphere (98% nitrogen). Following the heat treatment, the material was promptly placed inside a glovebox filled with argon (O₂ < 0.2 ppm, H₂O < 0.1 ppm) to avoid any exposure to moisture.

Materials characterizations

Powder X-ray diffraction (PXRD). Ex-situ lab-based PXRD measurements were performed on a Bruker D8 Advance diffractometer equipped with a Cu Kα X-ray source in Bragg-Brentano geometry. The measurements were recorded from 10° to 80° in the 2θ scan range with a step size of 0.02° and an exposure time of 0.8 s for each step. The current and voltage were set to 40 mA and 40 kV, respectively. Synchrotron PXRD was conducted at the BL17UM beamline using a large area detector (EIGER2-X-16M). The X-ray wavelength was $\lambda \approx 0.6209$

and 0.4838 Å. Samples were prepared by sealing the powder with Kapton tape inside an argon-filled glovebox to avoid air exposure. Rietveld refinements were performed using the TOPAS software (version 5)^{31,35,52}. The FAULTS program was utilized to simulate the stacking faults in the original NLMTO structure (Supplementary Table S3). The intensities of ex situ diffraction patterns have been normalized so that the highest intensity is set to 1.

Operando XRD was carried out on the same diffractometer equipped with Anton Paar XRK900 reactor chamber. The powder precursor mixture was prepared using the aforementioned method, and subsequently was pressed into pellets with a diameter of 10 mm under pressure of ~380 MPa. For the *operando* characterization of the Na-Mn-O system, Na₂O₂ and MnO₂ in a molar ratio of 3:4 was mixed by hand and pressed into pellets under pressure of ~380 MPa. The same precursor preparation method was employed for the *operando* synthesis of Li-Mn-O, using Li₂O and MnO₂ in a molar ratio of 1:4. Each sample loading was carried out inside a nitrogen-filled glove bag to prevent exposure to air. The reactor was purged using a 2% O₂ in N₂ flow at a rate of 50 sccm for 2 h before the calcination began. The sample was then heated from room temperature to 700 °C at a rate of 1 °C/min and maintained at 700 °C for 8 h under the 2% O₂ in N₂ flow (50 sccm), which scanned from 10° to 60° in the 2θ scan range with a step size of 0.02° and an exposure time of 0.6 s for each step. Each scan takes ~25 min and the interval between the end of a scan and the beginning of the next one is ~1 min.

Operando gas chromatography (GC). The home-built *operando* GC setup is mainly composed of, in sequence, a set of mass flow controllers (MFCs, Shaanxi Yidu Intelligence Technology Co., Ltd.), a tube furnace (TF5503SC-1, Thermo Fisher), a de-moisturizing column filled with silica gel and molecular sieves (4 Å) and a gas chromatograph equipped with a thermal conductivity detector (TCD) (9790Plus, Fuli Instruments). The column used was molecular sieve 5 Å, 60–80 mesh, 2 m × 1/8 inch. The GC is also equipped with a solenoid 6-way valve, thus allowing continuous sample injection and therefore *operando* measurement of the composition of the gas coming out from the tube furnace.

During the measurements, the column oven and TCD detector temperatures of the GC were set at 60 °C and 120 °C, respectively. Helium (Shanghai YouJiaLi Liquid Helium Co., Ltd. ≥99.999%) was used as carrier gas with a flow rate of 20 mL/min. Each scan takes ~2 min and the interval between the end of a scan and the beginning of the next one is ~0.2 min. The temporal resolution is sufficient for this work considering the temperature ramping section is typically ~11 h (1 °C/min rate). The instrument calibration was done by measuring gas flows containing 1%, 1.2%, 1.5%, 2%, and 5% oxygen in nitrogen, respectively, and an excellent linear correlation ($R^2 = 0.9999$) was obtained between the integral of the GC signal and the oxygen content (Supplementary Fig. 1). This calibration curve was used for processing the data from subsequent *operando* measurements. The response time (the time delay between the change in oxygen concentration in the reactor and the sampling in the GC) was estimated to be ~15 min, which is acceptable as the calcination experiment is significantly longer (Supplementary Fig. 1).

In a typically experiment, 5 mmol Na₂O₂ (or 10 mmol NaOH), 1.67 mmol Li₂O and 6.67 mmol MnO₂, targeting 10 mmol O3-NLMO, was mixed and loaded into a quartz tube (20 mm in diameter, 500 mm in length) in an argon glovebox. The quartz tube was sealed and transferred out from the glovebox, and then mounted onto the tube furnace with gas inlet and outlet connections. Subsequently, the whole setup was purged with the reaction gas at a flow rate of 100 sccm for 2 h prior to the calcination process. The reaction was conducted under the same gas environment and the same flow rate. The sample was heated from room temperature to 700 °C at a rate of 1 °C/min and maintained at 700 °C for 8 h.

The *operando* GC measurement gives a series of oxygen content values (in volume/molar percentage) as a function of time, denoted as $C_{O_2}(t)$ (%). The incoming gas has a total volume flow rate of 100 sccm, which is equivalent to a molar flow rate ($\dot{q}_n^*$) of 681.6 μmol/s at 298 K assuming the ideal gas condition. The following equation was used to calculate the net change in the molar flow rate of oxygen $\Delta q_{n,O_2}(t)$ at a given time t :

$$\Delta q_{n,O_2}(t) = \left(\frac{C_{O_2}(t) - C_{O_2}^0}{100 - C_{O_2}(t)} \right) \times \dot{q}_n^* (\mu\text{mol/s})$$

where $C_{O_2}^0$ (%) denotes the molar percentage of the oxygen in the gas flow going into the reactor. A positive $\Delta q_{n,O_2}(t)$ (μmol/s) value represents an increase in the oxygen flow rate, i.e., an oxygen-release reaction occurs in the reactor. The cumulative net change of oxygen was obtained by integrating the $\Delta q_{n,O_2}(t)$ vs t curve. The cumulative net change of oxygen (excluding the amount associated with the disproportionation of Na₂O₂) was further used to calculate the nominal Mn valance based on the assumption that Mn is the only element changes oxidation state during the reaction.

Thermogravimetry analysis. A TG 209 F3 Tarsus® instrument from NETZSCH was used for all the TGA experiments. The sample was prepared in an argon glovebox and then loaded into the TG (typically ~30 mg) using a nitrogen-filled glove bag (relative humidity ≤5%) to minimize moisture contamination from air. The powder samples were heated from room temperature to 700 °C with a 1 °C/min ramp-up rate under specific oxygen concentration atmosphere at a total flow rate of 100 sccm. It is worth noting that this instrument is configured so that the protection gas for the balance chamber (nitrogen) will enter the reaction chamber before venting. Therefore, the protection gas flow should be taken into account when setting up measurements targeting desired oxygen content for the reaction. All the gas supplies are controlled by built-in high precision mass flow controllers.

Operando mass spectrometry. The configuration of the *operando* mass spectrometry setup is similar to that of the *operando* GC with the only exception that a quadrupole mass spectrometer (HPR-20 EGA, Hiden) is used instead of a GC. The sample preparation was carried under the same routine as the *operando* GC experiments as introduced above. Gaseous species of N₂, O₂, H₂O, and CO₂ are monitored using the mass spectrometer, with ion peak channels set at 28, 32, 18, and 44, respectively. The setup was kept running for 10 h prior to the beginning of the reaction to achieve a stable signal baseline. During the temperature ramp, the powder sample was heated from room temperature to 700 °C at a rate of 1 °C/min using a 50 sccm flow of 2% O₂/98% N₂ gas until reaching 700 °C. The QGA software was used for quantitative analysis of the obtained results. Note that the mass spectrometer has not been calibrated so the results can only be used as a qualitative indicator.

Inductively coupled plasma-optical emission spectroscopy (ICP-OES). The chemical composition of the obtained O3-NLMO and O3-NLMTO materials was analyzed by inductively coupled plasma-optical emission spectroscopy (ICP-OES iCAP 7400). Sample were typically prepared by dissolving 20 mg of the cathode powder in 2 mL of aqua regia overnight. The solution was then filtered through nylon filter membrane and transferred to a 50 mL volumetric bottle.

Electron microscopy. The morphology and elemental distribution of the as-synthesized materials was examined by scanning electron microscopy (SEM, JSM-IT500HR/LA, JEOL) in conjunction with an energy dispersive X-ray spectroscopy (EDS, JEOL). For transmission electron microscopy (TEM) measurements, the samples were prepared by crushing the crystals with an agate mortar and pestle and dispersed

in alcohol by ultrasound. Multiple drops of suspension were deposited onto a copper-micro mesh grid. After drying, HAADF-STEM images and electron energy loss spectroscopy (EELS) were acquired on a probe aberration-corrected JEOL JEM-ARM300F transmission electron microscope operated at 300 kV.

X-ray absorption spectroscopy (XAS). Soft XAS experiments were conducted at the beamline 02B02 on the SiP-ME2 platform of the Shanghai Synchrotron Radiation Facility (SSRF)⁵³. The Mn L-edge spectra were acquired using the total electron yield (TEY) method and the beam size was configured to $150\ \mu\text{m} \times 50\ \mu\text{m}$ at the sample location. The measurements were performed at ambient temperature under ultra-high vacuum (below 1×10^{-9} Torr). All the spectra have been energy calibrated using the result of the reference sample MnO and normalized using the ATHENA software package⁵⁴. Mn K-edge XAS experiments were conducted at the BL16U1 beamline at SSRF. Data were acquired in transmission mode at room temperature. XAS data were processed and analyzed using the Demeter software package⁵⁴.

Vibrating Sample Magnetometer (VSM). Magnetic susceptibility measurements were carried out by VersaLab using a multi-function vibrating sample magnetometer (produced by Quantum Design), in the temperature range of 50–350 K and a magnetic field of 100 Oe. The sample was Ar protected to reacting with air. The results were fitted using the following equation based on a modified form of Curie-Weiss law to account for the temperature independent contribution to susceptibility: to fit the results, calculating the effective magnetic moment of the material:

$$\chi^{-1} = \frac{T - \theta_{\text{CW}}}{\chi_0 \times (T - \theta_{\text{CW}}) + C}$$

$$\mu_{\text{eff}} = \sqrt{8C} \mu_{\text{B}}$$

where T is the temperature, χ is the magnetic susceptibility, χ_0 is a magnetic factor independent of temperature, θ_{CW} represents the Curie-Weiss temperature, C is the Curie constant, μ_{eff} is the effective moment and μ_{B} is the Bohr magneton.

Batteries and electrochemistry. Self-standing cathode electrodes containing 75% (weight percentage) active material, 15% Super P conductive carbon and 10% polytetrafluoroethylene binder (PTFE, beads, Sigma-Aldrich) were prepared using the dry electrode processing method in a glovebox filled with argon ($\text{O}_2 < 0.2$ ppm, $\text{H}_2\text{O} < 0.1$ ppm)⁵⁵. Briefly, the as-synthesized cathode powder and Super P were hand-mixed in a mortar for a few minutes, followed by adding PTFE and repeated pressing and rolling using a stainless-steel rod (316 L; diameter: 10 mm) and hand pressure until a self-standing film was obtained. 10 mm disk electrodes with a typical mass loading of $\sim 8.6\ \text{mg cm}^{-2}$ and a thickness of around $40\ \mu\text{m}$ were punched out for cell assembly. The self-standing electrodes were assembled into 2032-coin cells (316 L, Canrd) with one piece of glass fiber (GF/B, Whatman) as the separator, and sodium metal (99.8%, Sigma-Aldrich) as the anode. All sodium metal electrodes were rolled by hand and stored in the glove box for less than one month and brushed with a toothbrush to remove any possible oxide layer on the surface before use. The sodium metal anode was typically $240\ \mu\text{m}$ in thickness ($\sim 25\ \text{mg cm}^{-2}$). The electrolyte used was 1 M NaClO_4 in ethylene carbonate/ethyl methyl carbonate 3/7 (EC/EMC) (Canrd) with 5% fluoroethylene carbonate (FEC, HSC Corporation; volume percentage). The electrolyte and its additives are commercial products that are stirred with teflon magnetons in a plastic bottle before assembling the battery. Constant-current cycling was performed on the LAND battery test system

(LANHE CT3002A, Wuhan LAND Electronics Co., Ltd.) between 1.5 and 4.5 V with two initial formation cycles at a rate of $0.05\ \text{C}$ ($9.5\ \text{mA g}^{-1}$), followed by aging cycles at a rate of $0.2\ \text{C}$ ($38\ \text{mA g}^{-1}$). All cells were kept at $25 \pm 1^\circ\text{C}$ in an ambient chamber (LBI-300HT, Shanghai Longyue Instrument Equipment Co., Ltd.) during the cycling. All battery testing experiments were conducted using at least 2–3 parallel cells. The specific capacity was obtained by normalizing the cell capacity by the mass of the active material.

Data availability

All data that support the findings in this study are available within this article and its Supplementary Information. Source data are provided with this paper. Any additional relevant data are available from the corresponding authors upon request. Source data are provided with this paper.

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Author contributions

C.X. conceived the idea. C.X., Z.L., and J.W. supervised the project. Y.Q., J.T., and Q.L. performed *operando* and ex situ PXRD and analysed data with assistance from F.L. W.W. and Y.W. assisted performing synchrotron X-ray diffraction. D.W., N.Y., and Y.Y. assisted the *operando* PXRD measurements. Y.Q. performed *operando* GC, TGA, SEM, XAS, electrochemistry experiments and analysed results with assistance from G.T. and N.Z. Q.L. performed VSM measurement, FAULTS analysis and theoretical calculations under guidance of J.W. J.T. performed *operando* MS and analysed data. P.Y. and K.C. conducted (S)TEM experiments and analysed data. C.X., Y.Q., and Q.L. wrote the manuscript with input from all co-authors.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Jingyang Wang, Zhengyan Lun or Chao Xu.

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